Differences in Resident Self-Evaluation and Clinical Competency Committee Evaluation Using ACGME Milestone Versions 1.0 and 2.0

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OBJECTIVE: Intentionally self-driven professional development of surgical resident physicians is a hallmark of surgical training and is expected to gain further traction as Entrustable Professional Activities (EPAs) become the new paradigm for surgical education. We aimed to analyze how surgical residents rate themselves as compared to the evaluation of the Clinical Competency Committee using ACGME Milestones Version 1 (M1.0) and Version 2 (M2.0).

DESIGN: We asked 22 general surgical trainees for self-evaluation of Milestones (both M1.0 and M2.0) from 2017 semiannually to 2022. ACGME-required Milestone evaluations by the Clinical Competency Committee (CCC) were independently performed after the time window for resident self-evaluation. Neither trainees nor CCC were aware of the other party's evaluations. There were 1552 paired data available for evaluating individual competencies by both trainees and CCC. Paired Wilcoxon signed-rank tests were then performed among the corresponding pairs.

SETTING: MercyOne Des Moines Medical Center, Des Moines, IA; Teaching tertiary referral center.

PARTICIPANTS: Twenty-two general surgical trainees at this hospital and 28 faculty surgeons participated in this study.

RESULTS: The average self-evaluation of surgical residents was lower in the M1.0 cohort compared to the corresponding CCC evaluation (1.96 ± 0.72 vs. 2.11 ± 0.67; $p < 0.001$). M1.0 self-assessments and CCC-assessments were statistically similar for ICS ($p = 0.548$) and PROF ($p = 0.554$) competencies and differed for MK ($p < 0.001$), PBLI ($p < 0.001$), PC ($p < 0.001$), SBP ($p = 0.008$). On the contrary, the M2.0 cohort demonstrated higher average self-evaluation of surgical residents compared to the corresponding CCC evaluation (2.75 ± 0.87 vs. 2.12 ± 0.97; $p < 0.001$). Significant differences were observed for all 6 ACGME competencies using M2.0 self-assessments and CCC-assessments (all $p < 0.001$). Multivariate regression modeling ($p < 0.001$, $R2 = 0.255$) predicted the degree of discordance between self-assessment and CCC-assessed achievement of competencies with a significant effect of gender (baseline male: coef $= -0.232$, $p < 0.001$), PGY level (-0.083 per year, $p < 0.001$) and Milestone version (0.831, $p < 0.001$). A significant interaction exists for all gender/Milestone combinations except for the female trainees with M1.0.

CONCLUSIONS: The difference between self-evaluated Milestone achievement and faculty-driven CCC evaluation of surgical resident physician performance is more evident in Milestones 2.0 than in Milestones 1.0. Residents self-evaluate higher compared to faculty using Milestones 2.0. This discrepancy is seen among both genders and is more pronounced among male residents overestimating core competencies with M2.0 self-evaluation than formal CCC assessment. (J Surg Ed 80:1378−1384. © 2023 Published by Elsevier Inc. on behalf of Association of Program Directors in Surgery.)

ABBREVIATIONS: CCC, Clinical Core Competency M1.0, Milestones version 1.0 M2.0, Milestones version 2.0 ICS, Interpersonal and Communication Skills MK, Medical Knowledge PBLI, Practice-Based Learning and Improvement PC, Patient Care PROF, Professionalism SBP,

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1931-7204/$30.00
https://doi.org/10.1016/j.jsurg.2023.07.005

Systems-based Practice EPAs Entrustable Professional Activities

KEY WORDS: milestones, core competencies, clinical competency committee, resident evaluation, gender-based discrepancies

COMPETENCIES: Patient Care, Medical Knowledge, Practice-Based Learning and Improvement, Interpersonal and Communication Skills, Professionalism, Systems-Based Practice

INTRODUCTION

The Accreditation Council for Graduate Medical Education (ACGME) and American Board of Medical Specialties (ABMS) introduced the 6 core competencies in 1999, which focused on the educational outcomes of training physicians. These competencies were found difficult to assess and underwent significant review to create narrative-based milestones that residents and faculty could more readily use to measure resident progression.[1] Milestones were implemented in 2013 and aimed to collect composite data, including resident self-evaluations, in order to assess progression towards competence before graduation and unsupervised practice.[2]

Feedback that Milestones version 1.0 (M1.0) used language that was too cumbersome led to the creation of the more streamlined Milestones version 2.0 (M2.0), which became effective on July 1, 2020 for General Surgery.[3] Multiple other specialties also noted that M2.0 were brief, straightforward, and better able to evaluate the integration of medical and non-medical care.[4,5]

There is an overall positive response towards the effects of milestones, though there are some discrepancies. Lyle et al.[6] described incongruities such as residents having a more negative self-evaluation compared to the Clinical Competency Committee (CCC) evaluation of residents when using M1.0. Female residents were also noted to have more negative self-evaluations than their male counterparts.

Given the previously described discrepancies between resident self-evaluation and CCC evaluation using M1.0, this study aimed to determine whether such discrepancies persist after the implementation of M2.0.

METHODS

This retrospective study was conducted on educational data between 2017 and 2022 in a community residency program. Milestones 1.0 was in place from 2017 to 2020, after which Milestones 2.0 was implemented. All general surgery resident physicians were evaluated by the Clinical Competency Committee (CCC) per contemporary ACGME policies. The program director instituted a mandatory concomitant self-evaluation of surgery residents with the intent to augment resident familiarity with ACGME competencies and to increase their professional self-reflection. In accordance with ACGME policies, milestones evaluations by the residents and the CCC were completed semi-annually at the mid-year and end of the year. Both evaluation sources were blinded to each other during the assessment period.

Because sub-competencies differ between the 2 versions of published Milestones, we collapsed individual sub-competencies into 6 corresponding core competencies (PC, MK, PBLI, ISC, PROF, SBC). The final analysis was thus performed on the 6 Core Competencies: Interpersonal and Communication Skills (ICS), Medical Knowledge (MK), Practice-Based Learning and Improvement (PBLI), Patient Care (PC), Professionalism (PROF), and Systems-based Practice (SBP).

Data were available from 22 surgery residents. Data from the Clinical Competency Committee was resulted from a total of 28 faculty staff surgeons, 4 of which were core members who were present throughout the study period, including the Program Director and Associate Program Director. Additional contribution led to a range of 6 to 14 total faculty members for each semi-annual CCC meeting. There were 1962 paired evaluations with 410 pairs missing (20%) from surgery residents, thus forming 1552 complete paired evaluations of individual sub-competencies.

Subgroup analysis was then conducted for the 6 Core Competencies to determine which categories were most at risk for discrepant evaluations. An additional analysis based on trainee gender was performed to elucidate gender-based differences.

The numerical values of achieved milestones in sub-competencies differ numerically between version 1.0 (range 0...4) and version 2.0 (1...5). To standardize the numerical difference, we defined evaluation variance (variance) as a difference between the self-evaluation of a resident physician and CCC evaluation for a given competency:

$$variance = evaluation^{self} - evaluation^{CCC}$$

Data are provided as mean ± standard deviation (median) for continuous variables. We used the Shapiro-Wilk test to assess the normality of distributions. We used variance testing between the groups and paired Wilcoxon signed-rank tests, as applicable for testing milestones. Between-group differences in educational variance were tested using the Mann-Whitney U test. $p < 0.05$ was considered statistically significant. Statistical analyses were performed using Stata 15 (Stata Corp, College Station, TX).

RESULTS

There were 1552 evaluations performed by the CCC paired with corresponding self-evaluations by surgical residents between 2017 and 2022. Thirteen female residents contributed 853 paired evaluations, and 9 male residents contributed 699 paired evaluations. A total of 564 evaluations were performed under Milestone 1.0 and 988 under Milestones 2.0 standards. Residents' compliance with self-evaluation was relatively high, with only 126 self-evaluations not completed, resulting in a compliance rate of 92.5%.

There was a strong correlation between CCC-evaluation and self-evaluation by residents under both Milestones versions (M1.0: $r = 0.648$, $p < 0.001$ and M2.0: $r = 0.691$, $p < 0.001$). The average self-evaluation of surgical residents was lower in the M1.0 cohort than the corresponding CCC evaluation (1.96 ± 0.72 vs. 2.11 ± 0.67; $p < 0.001$ by paired test; Fig. 1). On the contrary, the M2.0 cohort demonstrated higher average self-evaluation of surgical residents compared to the corresponding CCC evaluation (2.75 ± 0.87 vs. 2.12 ± 0.97; $p < 0.001$).

M1.0 self-assessments and CCC-assessments were statistically similar for ICS ($p = 0.548$) and PROF ($p = 0.554$) competencies and differed for MK ($p < 0.001$), PBLI ($p < 0.001$), PC ($p < 0.001$), SBP ($p = 0.008$). Significant differences were observed for all 6 ACGME competencies using M2.0 self-assessments and CCC-assessments (all $p < 0.001$; see Supplement).

We further analyzed differences in variance by gender. Under M1.0, both genders had a higher CCC evaluation score than their self-evaluation (males: 2.25 ± 0.61 vs. 2.10 ± 0.73 paired test $p < 0.001$, variance $p = 0.003$; females: 1.98 ± 0.70 vs. 1.83 ± 0.69, respectively). Conversely, under M2.0, both genders had a lower CCC evaluation score than their self-evaluation (males: 2.29 ± 0.99 vs. 3.07 ± 0.76 paired test $p < 0.001$, variance $p < 0.001$; females: 2.00 ± 0.93 vs. 2.50 ± 0.88, paired test $p < 0.001$, variance $p = 0.159$, respectively; Fig. 2).

To standardize the *a priori* dissimilarity between M1.0 and M2.0 scales, we examined evaluation variance as defined in Methods. Ideally, the variance between evaluation sources would be nil. We observed a significant change in variance between CCC evaluation and resident physician self-evaluations after converting from Milestones 1.0 to version 2.0 (Fig. 3). The difference between CCC evaluation and corresponding self-evaluation by resident physicians was similar between genders under M1.0 standards (males: -0.15 ± 0.58 vs. -0.14 ± 0.59 points among females, $p = 0.782$) but significantly different under M2.0 standards (males: $+0.78 \pm 0.66$ vs. $+0.50 \pm 0.76$ points among females, $p < 0.001$; Figs. 3A and B). As a group, male residents overestimated their competencies with M2.0 compared to the CCC assessment, a finding that was not identified among female residents.

Multivariate regression modeling ($p < 0.001$, $R_2 = 0.255$; Table) predicted the degree of discordance

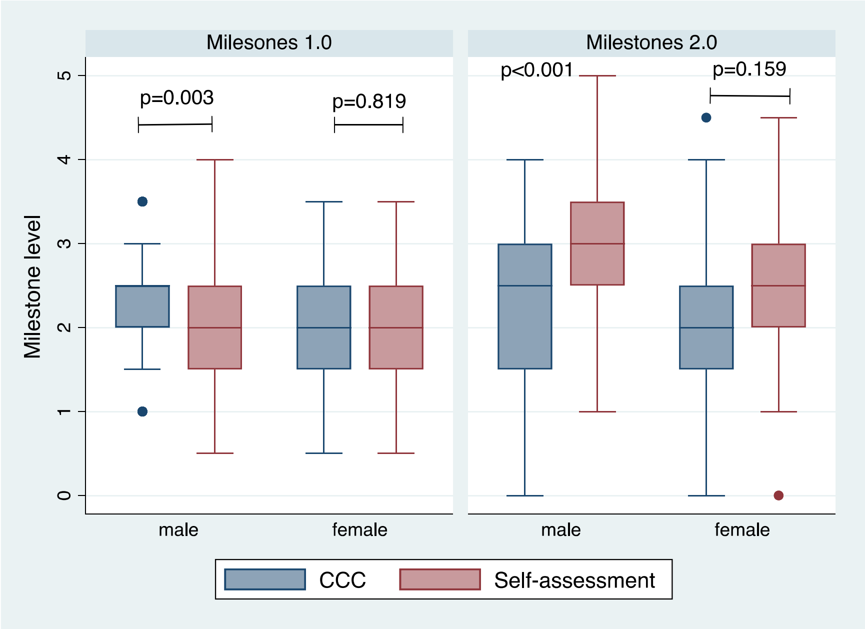

FIGURE 1. Pooled core competencies assessed by Clinical Competency Committee (CCC) contrasted to self-assessment by surgery resident physician. Statistical evaluation was conducted using paired Wilcoxon signed-rank test. While the range of milestone achievement is by definition numerically different between Milestones 1.0 (range 0...4) and 2.0 (range 1...5), the self-assessment appears substantially higher while using version 2.0.

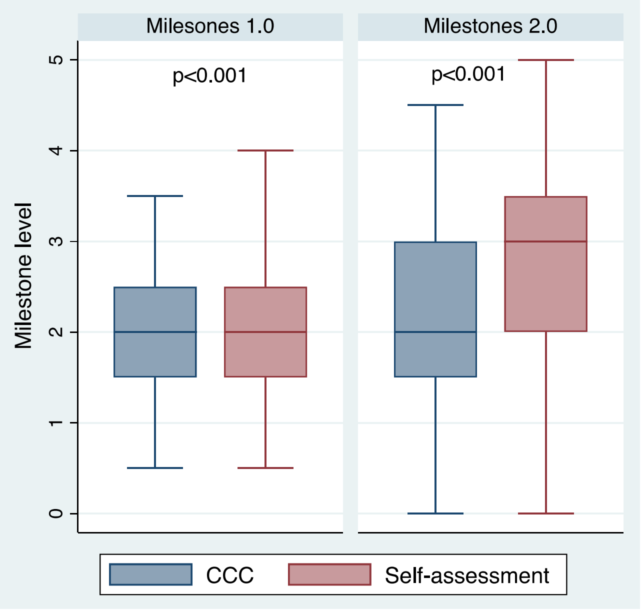

FIGURE 2. Core competencies assessed by Clinical Competency Committee (CCC) contrasted to self-assessment by surgery resident physicians by gender. Each pairwise comparison is significantly different ($p < 0.001$, not displayed). The displayed p-value is for between-group variance and appears mainly driven by males under Milestones 2.0.

between self-assessment and CCC-assessed achievement of competencies with a significant effect of gender. A significant interaction exists for all gender/Milestone combinations except for the female gender with M1.0.

DISCUSSION

The core competencies remain a foundation for evaluating resident progression throughout training.[7] Narrative milestones were created in 2013,[8] with a subsequent iteration for general surgery in 2019 leading to more specific and definitive guidelines to assess resident advancement.[9] Trainees' perceptions about their knowledge and skills differ with PGY level, gender, and other factors, such as the Dunning-Kruger effect.[6,10,11] Milestones 1.0 was studied to result in a more negative resident self-evaluation than the CCC assessment of the resident.[6] Our study, which evaluated such differences with the inclusion of the more recent Milestones 2.0, revealed significantly higher resident evaluation scores compared to that of the CCC, the effect of which is modified by trainee gender.

Explanations for these findings could be multifactorial. For one, M1.0 had a total of 16 sub-competencies[8] whereas M2.0 consisted of 18 sub-competencies,[9] which makes these tools inherently different in what they assess.

Moreover, major differences in how language is used between M1.0 and M2.0 make these assessment tools fundamentally different. M1.0 was criticized for having cumbersome language, which may have influenced residents' evaluations to be lower than the CCC's.[3,6] Many specialties noted this, and changes were made to limit the number of milestones for each sub-competency, decrease the number of words in each milestone, standardize certain milestones throughout all specialties, and modify the language to simplify it and make it more understandable.[4,5] In other words, the major changes in language and pursuit of a user-friendly format for M2.0 may have led to over-simplification of the description of certain competencies, leading to a higher resident self-evaluation compared to that of the CCC.

Another potential cause of the discrepancies could be that the composition of the CCC varied over time to some degree. However, the constant members of the CCC had considerable clinical experience, sustained their specific educational focus, and maintained their competency as educators. Ideally, evaluations from such experienced faculty and thoughtful resident self-evaluations would not be divergent.

Our 5-year study also noted significantly lower resident self-evaluations compared to the CCC for M1.0, which corresponds to the results seen by Lyle et al.[6] For M2.0, however, residents scored themselves higher than the CCC for all 6 core competencies. Given these findings, there may be an opportunity for improvement in milestone assessment, such as congruency between resident and CCC faculty evaluations. The use of Entrustable Professional Activities (EPAs) was noted to lead to equivalent resident self-evaluation and faculty evaluation scores.[12] Although EPAs led to mostly concordant evaluations, residents sometimes underestimated themselves as with Milestones 1.0, which may be desirable, as trainees might identify their own areas of improvement better than attending surgeons. This further urges us to improve Milestones 2.0 as it could lead to trainees overestimating themselves and forgoing opportunities for learning and self-improvement, which negatively impacts residents' long-term success.

There have been many gender-based analyses on resident evaluations with differing results. Some studies noted that female residents tended to be scored lower by faculty,[5,13] while other studies did not appreciate any differences in the male and female resident evaluations by the CCC.[14] Our study noted that there was no gender-based difference between trainee self-evaluation and CCC evaluation using M1.0. Conversely, using M2.0, male residents over-estimated themselves compared to the CCC. Female residents, on the other hand, did not significantly over-estimate themselves compared to the CCC using M2.0.

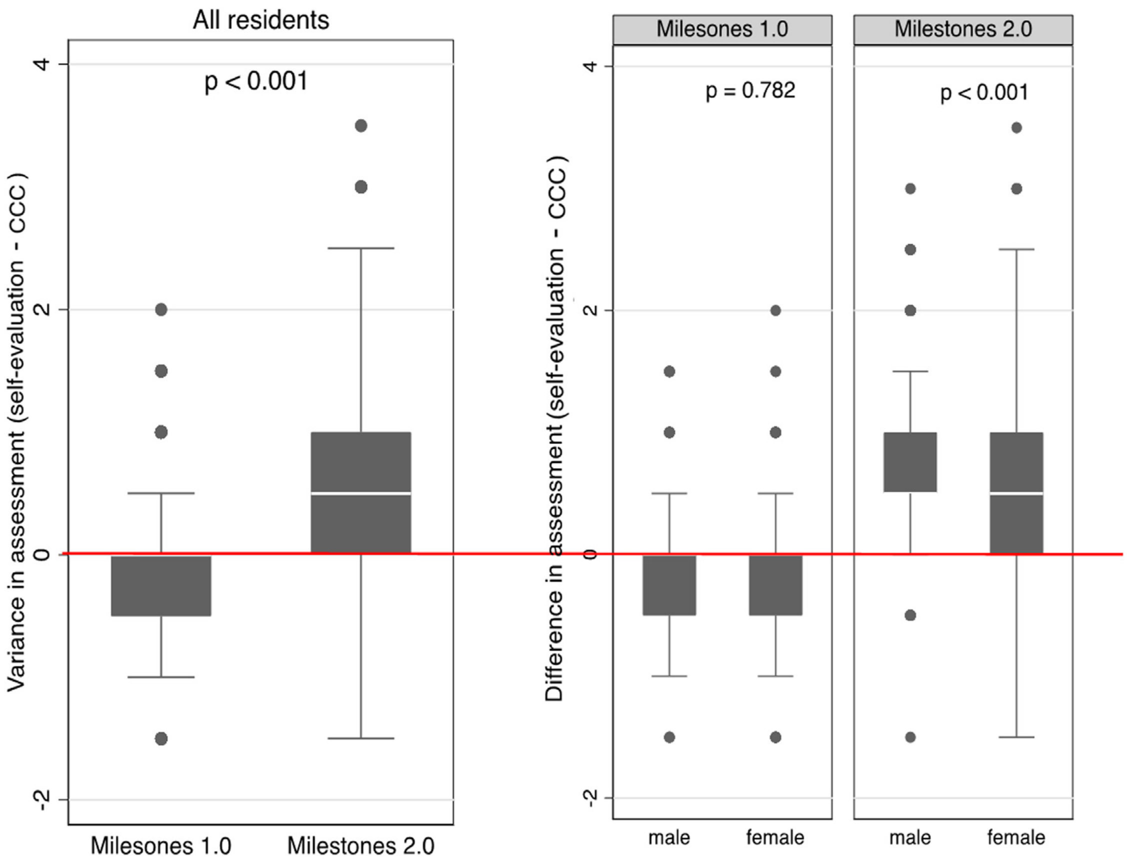

FIGURE 3. The variance between CCC and Self-Evaluation scores (pairwise mathematical difference self-evaluation − CCC scores). Left panel for all resident physicians based on Milestones 1.0 and 2.0. The same data is displayed by Milestone version and gender in 2 right panels. p-values are between-groups (not paired) differences by a Mann-Whitney test. The baseline hypothesis is that the variance between self-evaluation and CCC evaluation is zero (red line).

The reasons for these discrepancies could be multifactorial and may include implicit or explicit biases regarding trainee gender. It is well-studied that female residents tend to evaluate themselves as lower than male residents, but when it comes to autonomy, certain studies did not find a significant impact of gender on the meaningful autonomy a trainee received in the operating room.[15,16] Contrary to these findings, Meyerson et al[17] found decreased autonomy of female residents as evaluated by faculty and with resident self-evaluation. Qualitatively, written performance evaluations of male residents tended to be more positive regarding overall performance, future potential, and leadership skills.[18] Whether female residents actually receive equal autonomy may be controversial, but several studies support that female residents *perceive* a lower amount of autonomy, fewer opportunities, and tend to be less confident about their clinical skills.[19] Putting aside whether they have been just as autonomous and skilled or not, the findings seen with M2.0 show that females did not evaluate themselves significantly higher than the CCC as did their male counterparts. In other words, the use of M2.0 may be worsening the gender gap from M1.0 by decreasing faculty and resident self-perception of progression.

TABLE 1. Linear Regression Model Predicting the Difference Between CCC Evaluation and Self-Evaluation

Independent Variable	Coefficient	95% Confidence Interval	p-Value
Gender (male vs. female)	−0.232	−0.302, −0.162	<0.001
PGY level	−0.084	−0.113, −0.055	<0.001
Milestone Version (1.0 vs. 2.0)	0.832	0.760, 0.904	<0.001

The overall model provided excellent prediction, $p < 0.001$, $R2 = 0.255$. However, 75% variability was not explained by the independent variables. Baseline factor values are underlined

More studies are needed to evaluate how to better evaluate residents of all genders by eliminating bias. Implementing more objective tools such as EPAs may allow for significant decreases in gender biases when faculty assess male and female residents.[20] Faculty development, specifically on implicit bias training, may also help to mitigate such disparities in how trainees are perceived by faculty based on their gender. Future directions of study may utilize validated tools in assessing faculty implicit bias, length of experience of faculty, as well as resident evaluation of faculty, and how these may affect faculty evaluation of resident milestones.

There are several limitations of our study, including a small number of residents and the inability to concomitantly evaluate both M1.0 and M2.0 in the same period. This single-institution study may not be generalizable to other programs. Nevertheless, this is the first study comparing differences in resident self-evaluation and faculty evaluation of residents between 2 versions of Milestones.

CONCLUSIONS

The divergence between resident self-evaluation and Clinical Competency Committee evaluation has widened with adoption of Milestones 2.0. This discrepancy is mainly caused by higher self-evaluation among male residents.

DECLARATION OF COMPETING INTEREST

The authors declare that they have no conflicts of interest.

REFERENCES

1. Edgar L, Roberts S, Holmboe E. Milestones 2.0: A step forward. *J Grad Med Educ*. 2018;10(3):367–369. https://doi.org/10.4300/jgme-d-18-00372.1.
2. Nasca TJ, Philibert I, Brigham T, Flynn TC. The next GME accreditation system — rationale and Benefits. *N Engl J Med*. 2012;366(11):1051–1056. https://doi.org/10.1056/nejmsr1200117.
3. McLean S, Edgar L, Harsy B. The Milestones 2.0 Development Process. *J Grad Med Educ*. 2021;13(2s):4–7. https://doi.org/10.4300/jgme-d-20-00909.1.
4. Alaish SM, Arca MJ, Bucher BT, et al. Pediatric surgery milestones 2.0: A Primer. *J Pediatr Surg*. 2022;57(12):845–851. https://doi.org/10.1016/j.jpedsurg.2022.04.021.
5. Nayar R, Anderson S, Dyhdalo KS, et al. ACGME milestones 2.0: Why and what's new for cytopathology? *J Am Society Cytopathol*. 2021;10(5):485–494. https://doi.org/10.1016/j.jasc.2021.04.003.
6. Lyle B, Borgert AJ, Kallies KJ, Jarman BT. Do attending surgeons and residents see eye to eye? an evaluation of the Accreditation Council for Graduate Medical Education Milestones in general surgery residency. *J Surg Educ*. 2016;73(6):e54–e58. https://doi.org/10.1016/j.jsurg.2016.07.004.
7. Eno C, Correa R, Stewart NH, et al. Milestones guidebook for residents and fellows - ACGME. Available at: Milestones guidebook for residents and fellows. https://www.acgme.org/globalassets/PDFs/Milestones/MilestonesGuidebookforResidentsFellows.pdf. Accessed February 20, 2023. 2020.
8. The general surgery milestone project. *J Grad Med Educ*. 2014;6(1s1):320–328. https://doi.org/10.4300/jgme-06-01s1-40.1.
9. Surgery milestones - ACGME. Available at: https://www.acgme.org/globalassets/PDFs/Milestones/SurgeryMilestones.pdf. Accessed February 20, 2023.
10. Bilimoria KY, Quinn CM, Dahlke AR, et al. Use and underlying reasons for duty hour flexibility in the flexibility in duty hour requirements for surgical trainees (FIRST) trial. *J Am Coll Surg*. 2017;224(2):118–125. https://doi.org/10.1016/j.jamcollsurg.2016.10.046.
11. Rahmani M. Medical trainees and the dunning—kruger effect: When they don't know what they don't know. *J Grad Med Educ*. 2020;12(5):532–534. https://doi.org/10.4300/jgme-d-20-00134.1.
12. Stahl CC, Jung SA, Rosser AA, et al. Entrustable professional activities in general surgery: trends in resident self-assessment. *J Surg Educ*. 2020;77(6):1562–1567. https://doi.org/10.1016/j.jsurg.2020.05.005.
13. Anderson JE, Zern NK, Calhoun KE, Wood DE, Smith CA. Assessment of potential gender bias in general surgery resident milestone evaluations. *JAMA Surg*. 2022;157(12):1164–1166. https://doi.org/10.1001/jamasurg.2022.3929.
14. Kwasny L, Shebrain S, Munene G, Sawyer R. Is there a gender bias in milestones evaluations in general surgery residency training? *Am J Surg*. 2021;221(3):505–508. https://doi.org/10.1016/j.amjsurg.2020.12.020.
15. Lane SM, Young KA, Hayek SA, et al. Meaningful autonomy in general surgery training: exploring for gender bias. *Am J Surg*. 2020;219(2):240–244. https://doi.org/10.1016/j.amjsurg.2019.11.035.
16. Cookenmaster C, Shebrain S, Vos D, Munene G, Miller L, Sawyer R. Gender perception bias of operative autonomy evaluations among residents and faculty in general surgery training. *Am J Surg*. 2021;221(3):515–520. https://doi.org/10.1016/j.amjsurg.2020.11.016.

17. Meyerson SL, Odell DD, Zwischenberger JB, et al. The effect of gender on operative autonomy in general surgery residents. *Surgery*. 2019;166(5):738–743. https://doi.org/10.1016/j.surg.2019.06.006.

18. Gerull KM, Loe M, Seiler K, McAllister J, Salles A. Assessing gender bias in qualitative evaluations of surgical residents. *Am J Surg*. 2019;217(2):306–313. https://doi.org/10.1016/j.amjsurg.2018.09.029.

19. Nomura K, Yano E, Fukui T. Gender differences in clinical confidence: a nationwide survey of resident physicians in Japan. *Acad Med*. 2010;85(4):647–653. https://doi.org/10.1097/acm.0b013e3181d2a796.

20. Padilla EP, Stahl CC, Jung SA, et al. Gender differences in entrustable professional activity evaluations of general surgery residents. *Ann Surg*. 2021;275(2):222–229. https://doi.org/10.1097/sla.0000000000004905.

SUPPLEMENTARY INFORMATION

Supplementary material associated with this article can be found in the online version at doi:10.1016/j.jsurg.2023.07.005.